Task 1:

Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

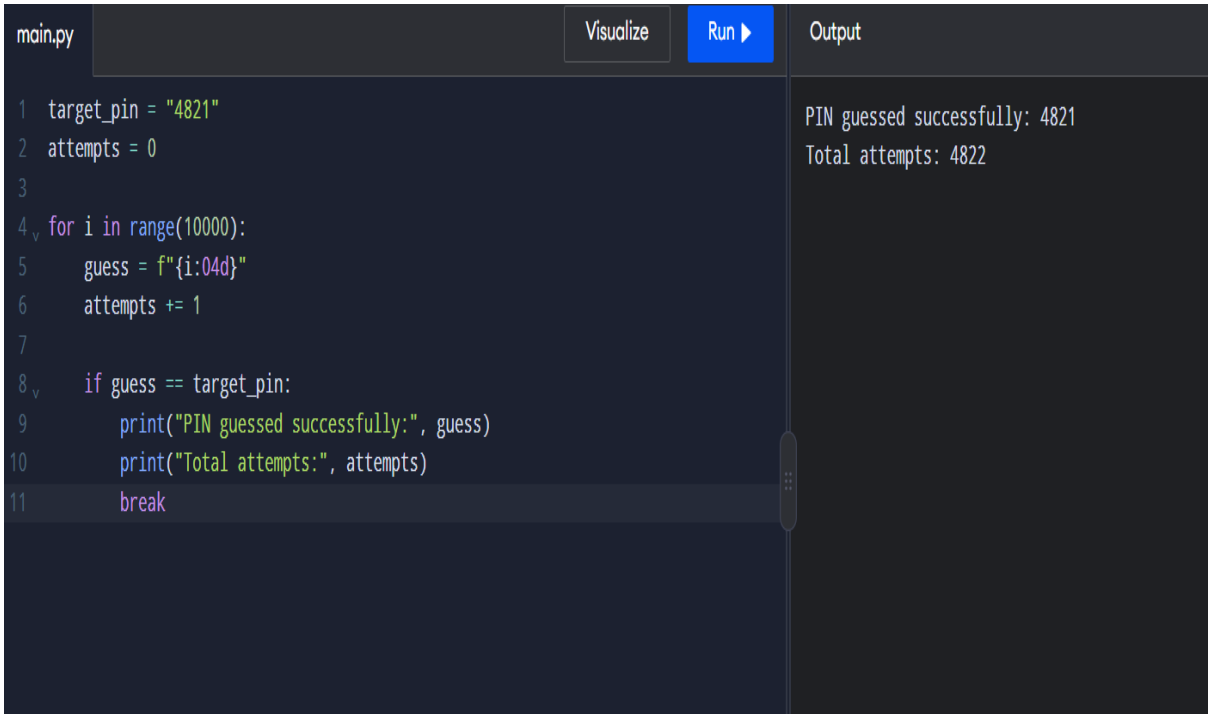
Answer:

No.	Network Threat	Real-World Example
1	Phishing	A fake Paytm message asks users to click a link and enter their OTP or banking details.
2	DDoS Attack	Attackers send massive amounts of fake traffic to a service, making an app such as Instagram unavailable.
3	Man-in-the-Middle (MitM)	An attacker on an unsecured public Wi-Fi network attempts to intercept users' communications.
4	Malware	A malicious APK pretending to be a WhatsApp modification app steals personal data.
5	Password Attack	An attacker repeatedly tries different passwords to gain unauthorized access to an Instagram account.

Task 2:

Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

Answer:



Task 3:

List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Answer:

No.	Physical Security Risk	Mitigation Step
1	Unauthorized access by intruders	Install biometric access control and use restricted entry cards.
2	Fire or overheating	Install fire suppression systems and proper cooling or air conditioning.
3	Power failure	Use a UPS and backup generator to maintain power.
4	Water leakage or flooding	Install water sensors and keep server equipment away from water sources.

Task 4:

Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

Answer:

Draw a simple office floor plan on paper and label the following areas:

- Entrance and reception
- Employee workstations
- Server room
- Meeting room
- Storage room
- Pantry and washroom

Mark these three security measures on your floor plan:

1. CCTV Camera: Near the entrance and server room.
2. Biometric Access Control: At the server room door.
3. Cable Locks: On server racks and network equipment.

Write a small legend below the drawing explaining each security measure.

Task 5:

Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

Answer:

A. General Checklist — 7 Actions

1. Keep operating systems, routers, firewalls, and applications updated.
2. Use strong passwords, multi-factor authentication, and role-based access control.
3. Monitor network traffic and logs for suspicious activities.
4. Regularly back up important data and network configurations.
5. Restrict physical access to servers and network equipment.
6. Train users to recognize phishing attacks and social engineering.
7. Prepare an incident response plan and test it regularly.

B. Modified Checklist for a College Computer Lab

1. Keep lab computers, switches, and routers updated with the latest security patches.
2. Use strong passwords and allow administrator access only to authorized staff.
3. Monitor the lab network for unusual traffic, port scanning, and malware.
4. Regularly back up important student records and network configurations.
5. Secure the computer lab with CCTV, locked doors, and cable locks for network devices.
6. Educate students about safe internet use, phishing links, and unknown downloads.
7. Maintain and periodically test an incident response plan for network attacks, theft, and equipment damage.